CYCLISTIC BIKE-SHARE

Case Study

How casual riders and annual members use Cyclistic differently, and how those differences can inform membership conversion strategy.

Prepared by Demitri Lee

6.17M	6.10M	12	5
RAW RIDES	CLEAN RIDES	MONTHS ANALYZED	CORE ANALYSES

Project Snapshot

Business objective: Increase annual memberships by better understanding how casual riders differ from annual members.

Primary analysis question: How do annual members and casual riders use Cyclistic bikes differently?

Analysis period: October 1, 2025 through September 30, 2026.

Tools: Google BigQuery, SQL, Google Colab, Python/pandas, and Google Sheets.

Data: Divvy historical trip data used for the Cyclistic case study, provided under license.

Executive Summary

Casual rides were more concentrated on weekends, in warmer months, at tourist and recreational locations, and were longer on average. Annual member rides were more evenly distributed across the workweek and showed stronger peaks during typical commuting hours. These patterns support targeted membership campaigns by location, ride behavior, and timing.

Data & Methodology

The analysis used 12 monthly Divvy historical trip-data CSV files for the Cyclistic case study. Each row represents a bike trip identified by a unique **ride_id**, with fields including bike type, timestamps, station information, coordinates, and rider type.

6,171,785	71,830	6,099,955
RAW ROWS	ROWS REMOVED	FINAL ROWS

Prepare

All 12 monthly files from October 2025 through September 2026 were reviewed for consistent structure, duplicate ride IDs, missing values in key fields, valid rider and bike categories, and unusual ride durations. The data was suitable for comparing casual and annual member trip behavior, but it does not identify individual riders.

Process

BigQuery and SQL were used to process the full 12-month dataset. The monthly files were uploaded through Google Colab and combined into a raw table while the original monthly tables were preserved.

Daylight-saving adjustment: 29 negative-duration rides occurred around 1:00 AM on November 2, 2025. Adding 60 minutes produced plausible durations of approximately 5 to 47 minutes, so these records were retained with corrected calculated durations.

Suspicious sub-one-minute rides: 159,192 rides were investigated. The final rule removed a sub-one-minute ride only when duration was 0 seconds, the start and end station were the same, or implied straight-line speed exceeded 50 km/h. This removed 66,506 rides while retaining 92,686.

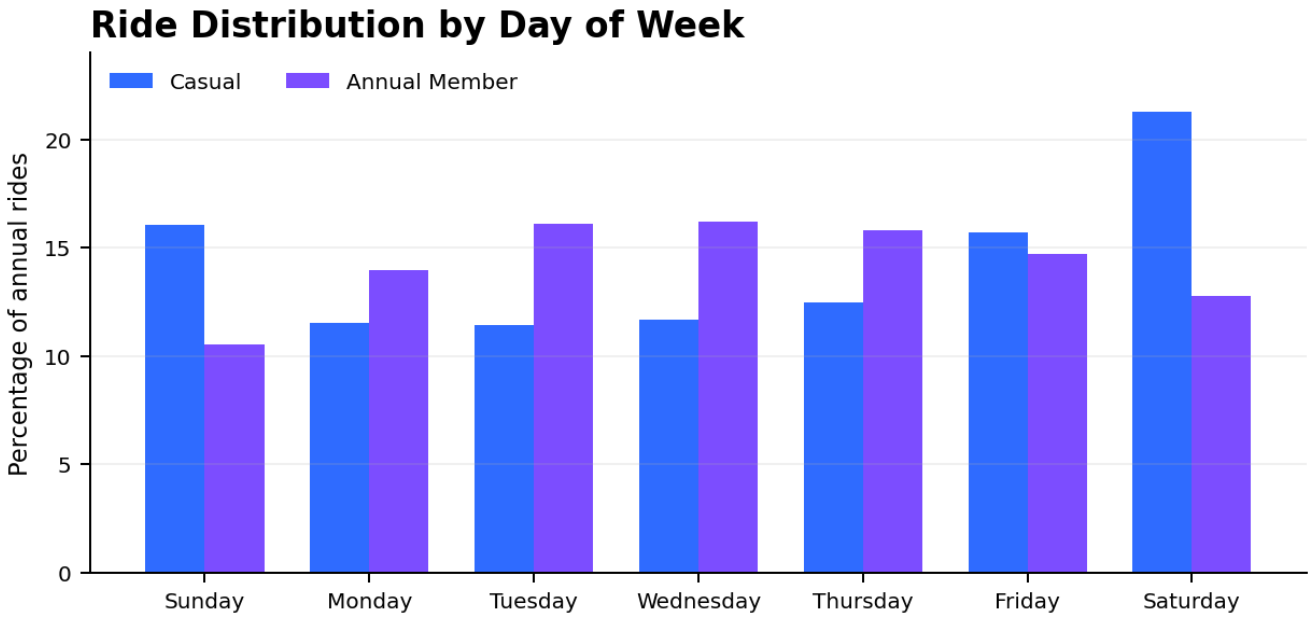
Rides over 24 hours: 5,324 rides were removed because of the highly consistent 24-to-25-hour pattern and incomplete ending information.

Final validation: The clean table had no negative calculated durations, no rides over 24 hours, no 0-second rides, no sub-one-minute same-station rides, and no sub-one-minute rides with implied speed greater than 50 km/h.

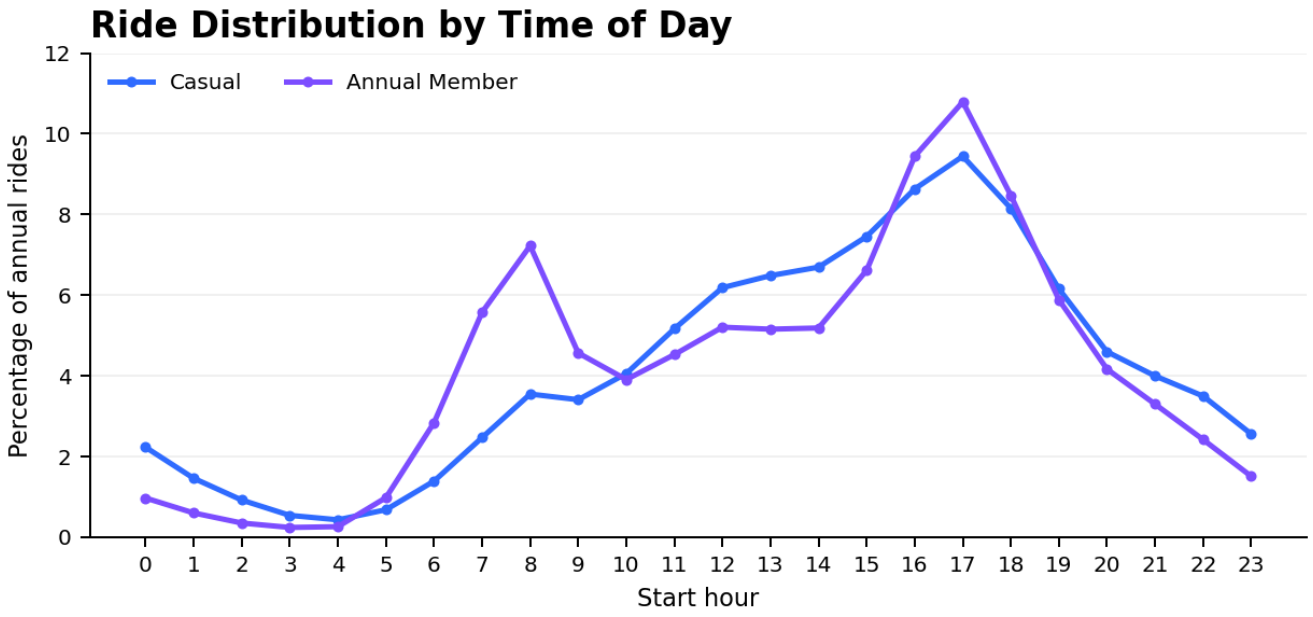
All final analysis queries were restricted to October 1, 2025 through September 30, 2026 to exclude any records falling outside the intended period.

Key Findings: When Riders Use Cyclistic

1. Casual rides are more weekend-focused. Saturday accounted for **21.25%** of all casual rides, compared with **12.76%** of annual member rides. Member rides were more evenly distributed across the workweek.

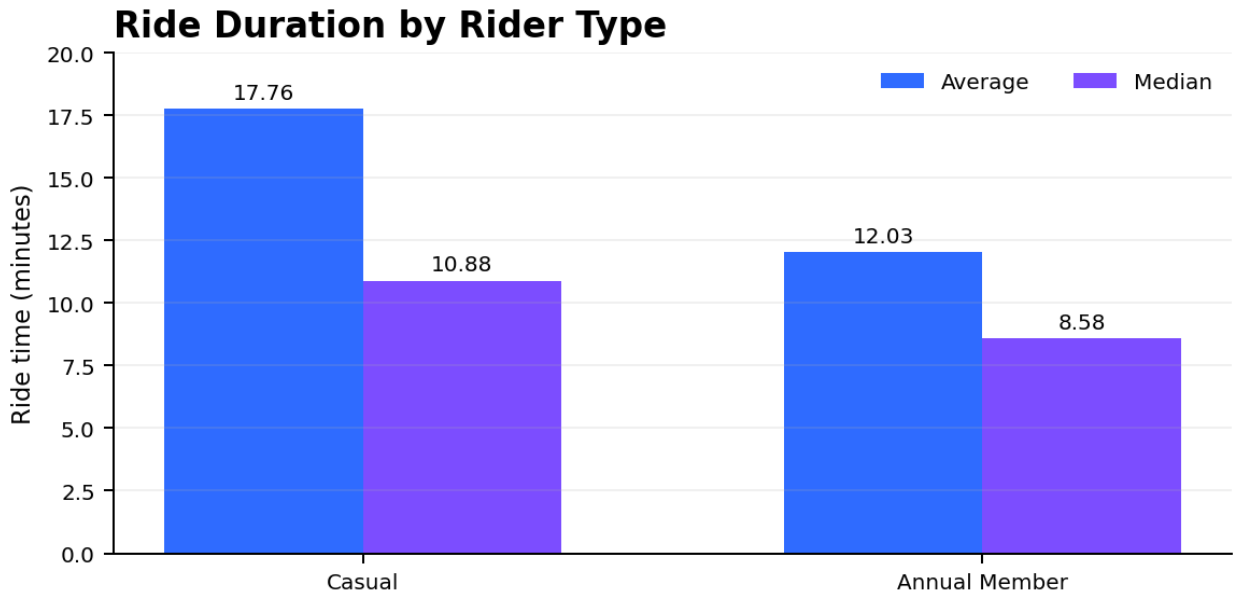


2. Annual members show stronger commuting-related timing. Member usage peaked at 7 to 8 a.m. and 4 to 6 p.m., with the largest share at **5 p.m. (10.79%)**. Casual usage built more gradually through the afternoon and early evening.

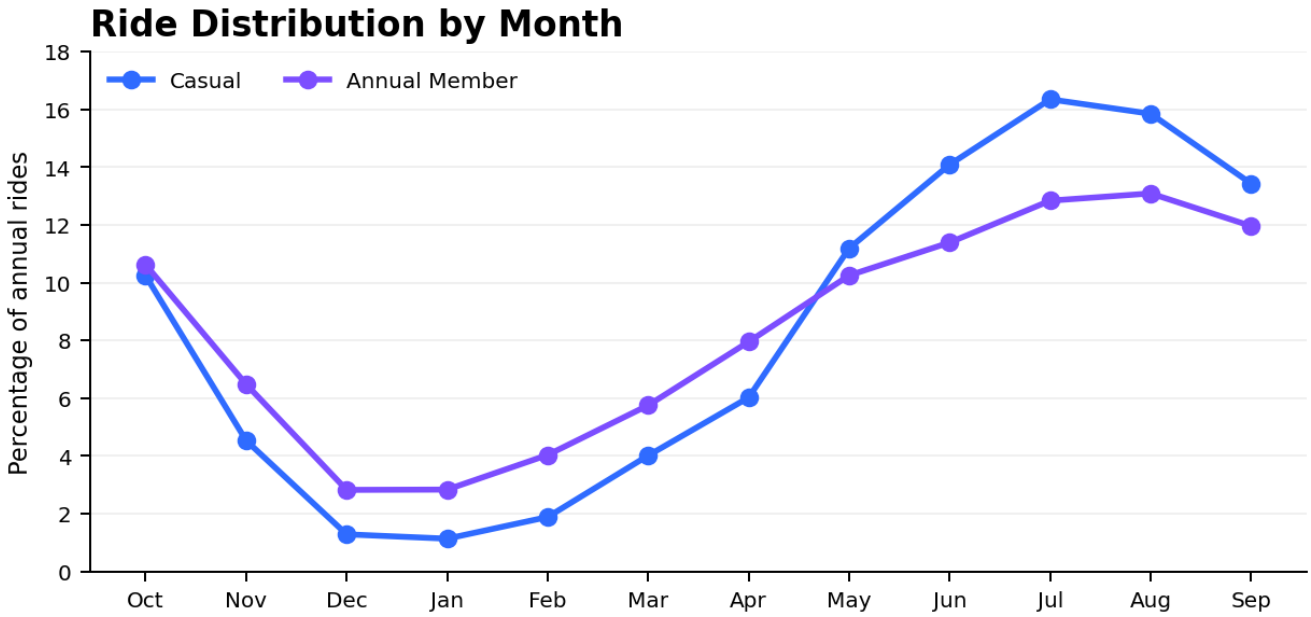


Key Findings: Ride Length & Seasonality

3. Casual rides tend to last longer. Casual rides averaged **17.76 minutes** versus **12.03 minutes** for annual members. Median duration was also higher for casual rides: **10.88 minutes** versus **8.58 minutes**.

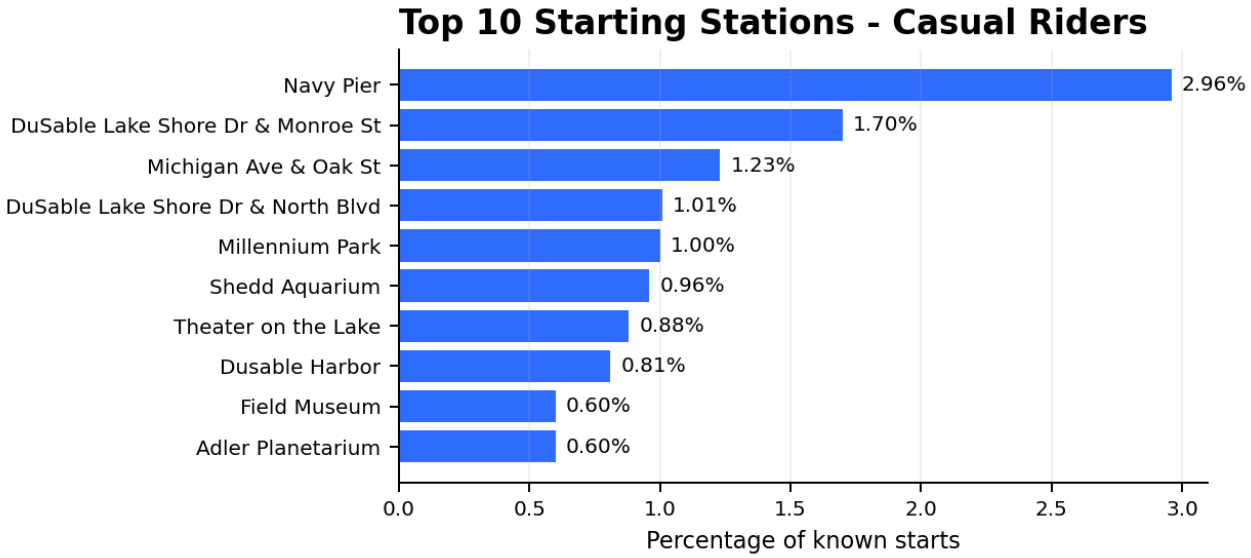


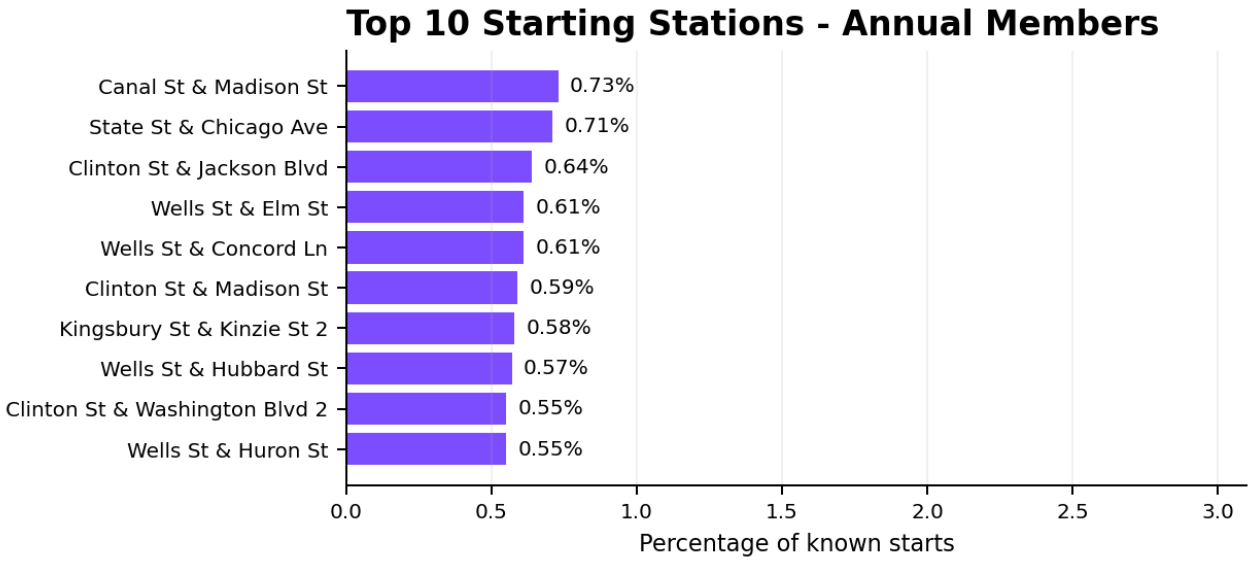
4. Casual usage is more seasonal. Both rider types were more active in warmer months, but casual rides showed larger seasonal swings. Casual rides reached their highest monthly share in **July (16.35%)**, while annual member rides peaked in **August (13.09%)**.



Key Finding: Where Rides Start

5. Casual rides are more concentrated near tourist and recreational destinations. Navy Pier alone accounted for **2.96%** of casual rides with known starting stations. The top 10 casual stations represented approximately **11.75%** of known casual starts, compared with **6.14%** of known member starts.





The station pattern suggests that casual riders may use Cyclistic more for leisure and recreational trips, while annual members appear more likely to use the service for routine transportation. Trip purpose is not directly observed in the data, so this interpretation remains directional rather than definitive.

Business Recommendations

1. Target high-traffic recreational and tourist stations

Promote annual membership at locations where casual rides are concentrated, especially the highest-traffic recreational and tourist-area stations. Messaging should focus on convenience, frequent access, and the value of using Cyclistic repeatedly.

Evidence: The top 10 casual starting stations account for approximately 11.75% of known casual starts, compared with 6.14% of known member starts.

Measure success: Use location-specific QR codes or promotional codes to track membership conversion at targeted stations versus comparable non-targeted stations.

2. Promote membership upgrades after longer casual rides

Offer a limited-time membership discount or credit after eligible longer casual rides. This targets riders who have already demonstrated meaningful engagement with the service and gives Cyclistic a direct conversion opportunity.

Evidence: Casual rides averaged 17.76 minutes versus 12.03 minutes for annual members.

Measure success: Track offer exposure, redemption, and conversion to annual membership, then compare with casual riders who did not receive the promotion.

3. Concentrate membership campaigns on weekends and warmer months

Run membership campaigns during the periods when casual riders are most active, using station signage, in-app messages, email, or post-ride communication.

Evidence: Saturday represented 21.25% of casual rides versus 12.76% of member rides, and casual activity was heavily concentrated from late spring through summer.

Measure success: Compare campaign engagement and conversion rates with similar periods before the campaign or with unexposed riders.

Limitations & Conclusion

Limitations

The trip dataset does not include personally identifiable rider information. As a result, the analysis cannot identify unique riders, determine whether the same casual rider completed multiple trips, determine whether casual riders live in the service area, or confirm whether an individual purchased multiple single-ride passes.

Trip purpose is not directly recorded. Commuting, tourism, and recreational use are inferred from timing, duration, and station patterns rather than confirmed directly.

Some rides contain missing station information. These rides were retained when they passed the cleaning criteria, but records with unknown starting stations were excluded from station-ranking analysis.

Conclusion

Cyclistic casual riders and annual members use the service differently in several consistent ways. Casual rides are more weekend-focused, more seasonal, longer on average, and more concentrated around recreational and tourist destinations. Annual member rides are more evenly distributed through the workweek and show stronger peaks during typical commuting hours.

Together, these findings support a focused conversion strategy that reaches casual riders in the places, moments, and usage contexts where they are already most engaged. The strongest next step would be to test these recommendations through controlled campaigns and measure actual annual membership conversion.

21.25%	17.76 min	16.35%
CASUAL RIDES ON SATURDAY	CASUAL AVG. DURATION	CASUAL JULY SHARE

Source: Divvy historical trip data used for the Cyclistic case study, October 2025 through September 2026. Analysis and recommendations are based on the final cleaned dataset of 6,099,955 rides.

Data source: https://divvybikes.com/system-data

Data license: https://divvybikes.com/data-license-agreement